EPIDEMIOLOGICAL MACHINE LEARNING • POPULATION SCREENING TRIAGE

When Training Labels Change Who Gets Screened

Temporal evaluation of diagnosis-based versus biomarker-augmented labels for diabetes screening in NHANES

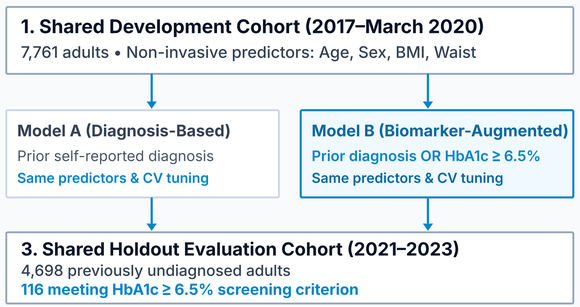
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THE CLINICAL PROBLEM

Population screening triage algorithms frequently train on self-reported prior diagnoses or electronic records. However, diagnosis-based labels can mark unrecorded disease negative when individuals face healthcare access barriers or lack recent clinical encounters.

STUDY DESIGN & TEMPORAL PROTOCOL

Evaluating whether augmenting diagnosis labels with venous HbA1c (≥ 6.5%) alters triage priorities in CDC NHANES, holding predictors and testing capacity constant:



METHODS & IMPLEMENTATION NOTE

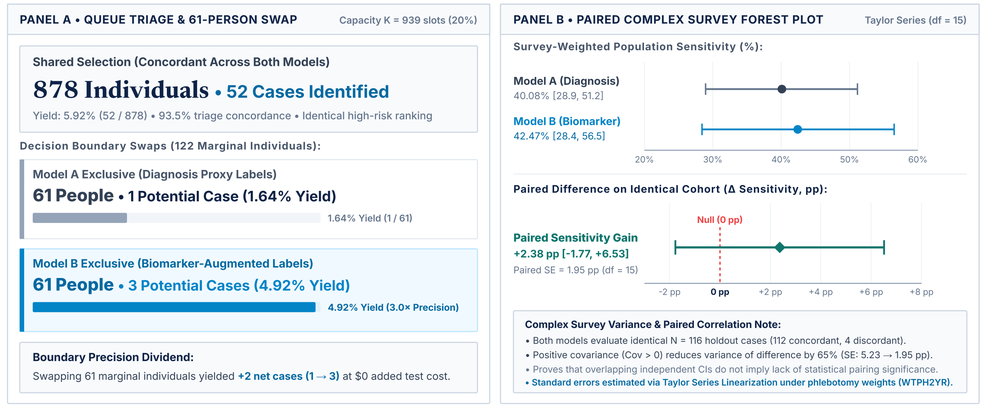
Primary models are **L2-regularized logistic regressions** tuned via 5-fold cross-validation on median-imputed, standard-scaled numeric features (age, BMI, waist circumference) and binary one-hot encoded sex. Poverty Income Ratio (PIR) and insurance were strictly excluded from model inputs and reserved solely for equity audits. Data pipeline ingested 10 XPT files across two survey periods using a reproducible DuckDB and Pandera pipeline that verified biological ranges (HbA1c: 2.0%–25.0%) and confirmed 0 pregnant participants.

KEY FINDING: **Retraining on observed venous biomonitoring alters screening prioritization and identifies two additional latent diabetes cases at identical 20% screening capacity.**

PAIRED-SELECTION QUEUE ANALYSIS (20% SCREENING CAPACITY)

Fixed Capacity K = 939 slots / 4,698 adults

Holding screening capacity fixed at 20% (939 test slots), both models prioritize 878 shared individuals. Priorities diverge for 122 individuals at the decision threshold:



Schematic: elements not to scale. Positive Predictive Value (PPV) is the percentage of screened individuals meeting the HbA1c criterion. The 3.0× yield difference (4.92% vs 1.64%) applies strictly to the 61-person boundary swap, not to overall model precision (5.86% vs 5.64% unweighted). Overall case detection increased from 53 to 55 (+2 potential cases).

BENCHMARK PERFORMANCE & COMPLEX SURVEY VARIANCE (20% CAPACITY • 939 SLOTS)

Screening Model	Unweighted Cases	Unweighted Sensitivity	Survey-Weighted Sensitivity (95% CI)	Survey-Weighted PPV (95% CI)
Model A (Diagnosis-Based)	53 / 116	45.69%	40.08% [28.9%, 51.2%]	5.79% [3.6%, 8.0%]
Model B (Biomarker-Augmented)	**55 / 116**	**47.41%**	**42.47% [28.4%, 56.5%]**	**6.02% [3.4%, 8.6%]**
Net Change (Δ)	**+2 Cases**	**+1.72 pp**	**+2.38 pp** [paired CI: -1.77, +6.53]	**+0.22 pp** (overlapping CIs)

Survey inference note: Weighted estimates use Taylor Series Linearization (WTPH2YR weights, df = 15). Paired survey analysis on identical N = 116 holdout cases reduces standard error to 1.95 pp (+2.38 pp sensitivity difference, paired 95% CI: [-1.77 pp, +6.53 pp]). Overlapping separate marginal CIs reflect small cluster degrees of freedom.

HEALTH EQUITY SUBGROUP AUDIT

Evaluating detection across insurance status at 20% budget:

Subgroup	Total Cases	Model A Detection	Model B Detection
Insured Adults	102 cases	51 / 102 (50.0%)	52 / 102 (51.0%)
Uninsured Adults	**13 cases**	**2 / 13 (15.4%)**	**3 / 13 (23.1%)**

Small denominator note: The uninsured cohort contains only **13 participants meeting the HbA1c screening criterion**; detection changed by +1 case (2/13 to 3/13). While directionally consistent with hypothesis, small cell counts preclude claims of disparity elimination or causal improvement (1 participant has unrecorded insurance coverage; total holdout cases = 116).

MULTI-ARCHITECTURE ROBUSTNESS

Triage re-evaluated under alternative tree architectures:

- **Random Forest:** Model A detects 60 cases; Model B detects 61 cases (+1 case, 77 swaps).
- **HistGradientBoosting:** Model A detects 59 cases; Model B detects 59 cases (0 net gain, 132 swaps).

Confirms priority shifts persist across architectures (+0 to +2 cases, up to 132 queue swaps). Linear logistic regression remains the primary reference model for clinical transparency, monotonicity, and guideline compliance.

LIMITATIONS & SCOPE

- **Surveillance vs Diagnosis:** Single venous HbA1c ≥ 6.5% represents an elevated screening criterion, not an autonomous clinical diagnosis.
- **Inference Bounds:** Paired survey sensitivity difference is +2.38 pp (paired 95% CI: [-1.77 pp, +6.53 pp]); overlapping separate marginal CIs reflect small holdout cluster degrees of freedom.
- **Operational Scope:** Assumes 100% screening adherence among prioritized adults and uniform laboratory costs.

CONCLUSION & NEXT STEPS

Augmenting administrative diagnosis labels with objective biomarker-augmented labels modestly refines borderline triage (+2 cases, 53 → 55) without expanding predictor requirements. Next steps require pooling additional survey cycles to reduce cluster-level variance (paired 95% CI: [-1.77, +6.53 pp]) and evaluating multimodal glycemic outcomes (fasting plasma glucose and OGTT) across diverse clinical settings.

References: American Diabetes Association. (2024). *Diabetes Care*, 47(Suppl. 1), S20–S42 • CDC NCHS. (2024). *NHANES 2017–2023*. U.S. DHHS • Obermeyer et al. (2019). *Science*, 366(6464), 447–453

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